

VOLUME OF A SMALL EXTRINSIC BALL IN A SUBMANIFOLD

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ABSTRACT

For a submanifold $M^p \hookrightarrow \mathbb{R}^N$, we determine a two-term asymptotic formula for $\text{vol}(M^p \cap B_\varepsilon(x))$ for $x \in M^p$ as $\varepsilon \downarrow 0$. The second term is a quadratic curvature invariant of the second fundamental form of the imbedding. Imbedded spheres are characterized among compact hypersurfaces by this term.

1. Introduction

In recent years there have been a number of works dealing with asymptotic results in differential geometry in which curvature invariants appear as coefficients in the expansions. Historically the first such formula is due to Bertrand, Diguët and Puiseux [1], who expressed the area of a non-Euclidean disk in terms of the curvature, to first order when the radius of the disk tends to zero. This formula was later generalized to geodesic balls in arbitrary Riemannian manifolds by Cartan [2] with subsequent improvements by Gray [3] and Gray and Vanhecke [5]. These papers yield characterizations of Euclidean space and other model spaces in terms of the volume of small geodesic balls.

In order to obtain more effective characterizations of Euclidean space and other model spaces, several papers have been devoted to the mean exit time of Brownian motion [4, 6, 8]. The asymptotic formulas for the mean exit time involve new quadratic curvature invariants. In the case of the mean exit time from a tubular neighborhood, one obtains a quadratic form in the principal curvatures whose lowest eigenvalue is simple and corresponds to *umbilic* submanifolds. In the case of *extrinsic balls* of an immersed manifold, the mean exit time is expressed in terms of the *mean curvature*, to the first order of asymptotics. This leads to a characterization of minimal hypersurfaces in terms of the mean exit time of Brownian motion.

In this paper we consider a submanifold of $\mathbb{R}^n$ and its intersection with a small ball of $\mathbb{R}^n$. The volume of the resulting intersection is computed in the asymptotic limit of radius tending to zero. This leads to a new asymptotic formula for the volume, in which appears a quadratic form which generalizes that obtained in studying the mean exit time of Brownian motion from a tubular neighborhood of a hypersurface ([6, Theorem 11]). In Section 3 we obtain some characterizations of spheres and totally geodesic imbeddings by means of the volume.

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2. Statement and solution of problem

Let M^p be a p -dimensional submanifold of $\mathbb{R}^N$. For $x_0 \in M^p$ we consider the intersection $M_\varepsilon^p(x_0)$ of M^p with the ambient space ball

$$B_\varepsilon(x_0) = \{y \in \mathbb{R}^N : |y - x_0| < \varepsilon\}$$

or equivalently $M_\varepsilon(x_0) = M^p \cap B_\varepsilon(x_0)$. The asymptotics of the volume of $M_\varepsilon(x_0)$ as $\varepsilon \downarrow 0$ are governed by the extrinsic curvatures of M^p as derived from the second fundamental form B , which is defined by the relation

$$B_{x_0}(v, w) = (\nabla_v W)^\perp \quad v, w \in T_{x_0} M,$$

where V and W are smooth extensions of v and w respectively, ∇ is the standard Levi-Civita connection of Euclidean space $\mathbb{R}^N$ and $\perp$ denotes the component normal to $T_{x_0} M$. The *mean curvature normal* is the vector

$$H_{x_0} = \text{trace } B_{x_0} = \sum_{i=1}^p B_{x_0}(e_i, e_i)$$

where $\{e_i\}$ is an orthonormal frame for $T_{x_0} M$; note that we have not divided by the dimension of M . Finally $\|H\|$ denotes the length of H and $\|B\|$ denotes the standard Hilbert-Schmidt norm of B : $\|B\|^2 = \sum_{ij} \|B(e_i, e_j)\|^2$, with $\{e_i\}$ as above. If M^n is a hypersurface in $\mathbb{R}^{n+1}$ with principal curvatures $\{\lambda_i\}$ then $\|B\|^2 = \sum \lambda_i^2$ and $\|H\|^2 = (\sum \lambda_i)^2$. For background concerning B , consult [7].

Having formulated these preliminaries, we can state the main result.

THEOREM. *The volume of the extrinsic neighborhood $M_\varepsilon^p(x)$ has the asymptotic behavior when $\varepsilon \downarrow 0$:*

$$\text{vol}(M_\varepsilon^p(x)) = (\sigma_{p-1} \varepsilon^p / p) + (\sigma_{p-1} / 8p(p+2)) \{2\|B\|_x^2 - \|H\|_x^2\} \varepsilon^{p+2} + O(\varepsilon^{p+3})$$

where σ_{p-1} is the surface measure of the unit sphere in $\mathbb{R}^p$.

Proof. We consider the imbedded p -dimensional submanifold $M^p \subset \mathbb{R}^N$ to be defined in a local representation by the smooth functions

$$x_{p+1} = \psi_1(x), \dots, x_N = \psi_q(x) \quad (2.1)$$

with $x = (x_1, \dots, x_p)$ being local coordinates in the tangent space TM at $x_0 \in M$, $x_0 = (0, \dots, 0)$ and $q = N - p$, the codimension of M^p . We have

$$M_\varepsilon^p = \{(x_1, \dots, x_N) : \sum_{i=1}^N x_i^2 \leq \varepsilon^2, (x_1, \dots, x_N) \in M^p\}.$$

Let $X(u) = (u, \psi_1(u), \dots, \psi_q(u))$. Then M_ε^p is the diffeomorphic image of the region $D_\varepsilon = \{u \in \mathbb{R}^p : |X(u)|^2 \leq \varepsilon^2\}$. The functions ψ_α may be written in the form

$$\psi_\alpha(x) = \frac{1}{2} \sum_{i,j} A_\alpha^{ki} x_k x_i + O(|x|^3) \quad (|x| \rightarrow 0) \quad (2.2)$$

where A_α^{ki} is symmetric in the upper indices. The volume is written as

$$\text{vol}(M_\varepsilon^p) = \int_{D_\varepsilon} \sqrt{g} \, du \quad g = \det(g_{ij})$$

where (g_{ij}) may be computed as

$$\begin{aligned}
 g_{ij} &= X_i X_j \\
 &= \delta_{ij} + \sum_{\alpha} \psi_{\alpha i} \psi_{\alpha j} \quad (\psi_{\alpha i} = (\partial \psi_{\alpha} / \partial u_i)) \\
 &= \delta_{ij} + \frac{1}{4} \sum_{\alpha k l m n} [A_{\alpha}^{kl} (\delta_{ki} x_l + \delta_{li} x_k)] [A_{\alpha}^{mn} (\delta_{mj} x_n + \delta_{jn} x_m)] + O(|x|^3) \\
 &= \delta_{ij} + \frac{1}{4} \sum_{\alpha} \left\{ \sum_l A_{\alpha}^{il} x_l + \sum_k A_{\alpha}^{ki} x_k \right\} \left\{ \sum_m A_{\alpha}^{jm} x_m + A_{\alpha}^{ni} x_n \right\} + O(|x|^3) \\
 &= \delta_{ij} + \sum_{\alpha} \left(\sum_l A_{\alpha}^{il} x_l \right) \left(\sum_k A_{\alpha}^{jk} x_k \right) + O(|x|^3) \\
 &= \delta_{ij} + \sum_{\alpha} \sum_{kl} B_{\alpha ij}^{kl} x_k x_l + O(|x|^3) \\
 &= \delta_{ij} + \sum_{\alpha} B_{\alpha ij}(\xi) r^2 + O(|x|^3)
 \end{aligned}$$

where

$$B_{\alpha ij}^{kl} = A_{\alpha}^{il} A_{\alpha}^{jk} \text{ and } B_{\alpha ij}(\xi) = \sum_{kl} B_{\alpha ij}^{kl} \xi_k \xi_l.$$

Thus

$$\sqrt{g} = 1 + \frac{1}{2} (B_{\alpha ii}(\xi)) r^2 + O(r^3).$$

Now the boundary ∂D_{ε} is defined by $\partial D_{\varepsilon} = \{u : |X(u)|^2 = \varepsilon^2\}$, or

$$\begin{aligned}
 \varepsilon^2 &= r^2 + \sum_{\alpha} \psi_{\alpha}(u)^2 \\
 &= r^2 + \frac{1}{4} \sum_{\alpha} \left(\left\{ \sum_{ij} A_{\alpha}^{ij} \xi_i \xi_j \right\} r^2 + O(r^3) \right)^2 \\
 &= r^2 + \frac{1}{4} Q r^4 + O(r^5)
 \end{aligned}$$

where $Q =: \sum_{\alpha} (\sum_{ij} A_{\alpha}^{ij} \xi_i \xi_j)^2$. Solving this quadratic equation for r^2 yields

$$\begin{aligned}
 r^2 &= (-1 + \sqrt{(1 - 4(\frac{1}{4}Q))((Or^5) - \varepsilon^2)}) / 2(\frac{1}{4}Q) \\
 &= (-1 + \sqrt{(1 + Q\varepsilon^2 + O(\varepsilon^5))}) / \frac{1}{2}Q \\
 &= (-1 + (1 + \frac{1}{2}Q\varepsilon^2 - \frac{1}{8}Q^2\varepsilon^4 + O(\varepsilon^5))) / \frac{1}{2}Q \\
 &= \varepsilon^2 - \frac{1}{4}Q\varepsilon^4 + O(\varepsilon^5)
 \end{aligned}$$

and

$$\begin{aligned}
 r &= \varepsilon(1 - \frac{1}{8}Q\varepsilon^2 + O(\varepsilon^3)) \\
 &= \varepsilon - \frac{1}{8}Q\varepsilon^3 + O(\varepsilon^4) =: h_{\varepsilon}(\xi).
 \end{aligned}$$

Consequently,

$$\begin{aligned}
 \text{vol}(M_{\varepsilon}^p) &= \iint_{D_{\varepsilon}} \sqrt{g} \, du \\
 &= \int_{S^{p-1}} d\xi \int_0^{h_{\varepsilon}(\xi)} [1 + \frac{1}{2} \sum_{\alpha} B_{\alpha ii}(\xi) r^2 + O(r^3)] r^{p-1} \, dr \\
 &= \int_{S^{p-1}} d\xi \int_0^{h_{\varepsilon}(\xi)} [r^{p-1} + \frac{1}{2} \sum_{\alpha} \dot{A}_{\alpha ii}(\xi) r^{p+1} + O(r^{p+2})] \, dr \\
 &= \text{I} + \text{II} + O(r^{p+3}),
 \end{aligned}$$

where

$$\begin{aligned}
 I &= : \int_{S^{p-1}} d\xi \int_0^{h_\epsilon(\xi)} r^{p-1} dr = (1/p) \int_{S^{p-1}} h_\epsilon(\xi)^p \\
 &= (1/p) \int_{S^{p-1}} d\xi [1 - \frac{1}{8}pQ\epsilon^2 + O(\epsilon^3)] \epsilon^p \\
 &= (\sigma_{p-1}/p) \epsilon^p - \frac{1}{8} \left[\int_{S^{p-1}} Q d\xi \right] \epsilon^{p+2} + O(\epsilon^{p+2}), \\
 II &= : \frac{1}{2} \int_{S^{p-1}} d\xi \int_0^{h_\epsilon(\xi)} \sum_{\alpha i} B_{\alpha i i}(\xi) r^{p+1} dr \\
 &= \frac{1}{2} \int_{S^{p-1}} d\xi \{ \sum_{\alpha i} B_{\alpha i i}(\xi) \} \epsilon^{p+2}/(p+2) + O(\epsilon^{p+3}).
 \end{aligned}$$

Therefore

$$\text{vol}(M_\epsilon^p) = (\sigma_{p-1}/p) \epsilon^p + \frac{1}{2} \epsilon^{p+2} \left\{ (p+2)^{-1} \sum_{\alpha i} \int_{S^{p-1}} B_{\alpha i i}(\xi) d\xi - \frac{1}{4} \int_{S^{p-1}} Q \right\} + O(\epsilon^{p+2}).$$

Now

$$\int_{S^{p-1}} B_{\alpha i i}(\xi) d\xi = \int_{S^{p-1}} \sum_{kl} B_{\alpha i i}^{kl} \xi_k \xi_l d\xi = (\sigma_{p-1}/p) \sum_k B_{\alpha i i}^{kk},$$

so that

$$\frac{1}{p+2} \sum_{\alpha i} \int_{S^{p-1}} B_{\alpha i i}(\xi) d\xi = \frac{\sigma_{p-1}}{p(p+2)} \sum_{\alpha i k} (A_\alpha^{ik})^2.$$

Now we make use of the identities $\int \xi_i^2 \xi_j^2 d\xi = \sigma_{p-1}/p(p+2)$ for $i-j$ non-zero and $\int \xi_i^4 d\xi = 3\sigma_{p-1}/p(p+2)$ while $\int H(\xi) d\xi = 0$ for all other homogeneous polynomials H of degree 4. This yields

$$\begin{aligned}
 \int_{S^{p-1}} Q d\xi &= \sum_\alpha \int_{S^{p-1}} \{ \sum_{ij} A_\alpha^{ij} \xi_i \xi_j \}^2 d\xi \\
 &= \sum_\alpha \int_{S^{p-1}} \sum_{ijkl} (A_\alpha^{ij} \xi_i \xi_j A_\alpha^{kl} \xi_k \xi_l) d\xi \\
 &= \sum_\alpha \int_{S^{p-1}} \{ \sum_{ijkl} A_\alpha^{ij} A_\alpha^{kl} \xi_i \xi_j \xi_k \xi_l \} d\xi \\
 &= \frac{2\sigma_{p-1}}{p(p+2)} \sum_\alpha \{ \sum_{ij} (A_\alpha^{ij})^2 + \frac{1}{2} \sum_i (A_\alpha^{ii})^2 \}.
 \end{aligned}$$

Combining these and substituting in the formula for volume, we have

$$\text{vol}(M_\epsilon^p) = \frac{\epsilon^p \sigma_{p-1}}{p} + \frac{\sigma_{p-1} \epsilon^{p+2}}{8p(p+2)} \{ 2 \sum_{\alpha ij} (A_\alpha^{ij})^2 - \sum_\alpha (\sum_i A_\alpha^{ii})^2 \} + O(\epsilon^{p+3})$$

where σ_{p-1} is the surface area of S^{p-1} . This completes the proof. In fact, it is a simple matter to check that, since the frame $\{\partial/\partial u_i\}$ is orthonormal at $u=0$, $B_{x_0} = (\text{Hess } \psi_\alpha)_0$, $\|B\|^2 = \sum (\partial^2 \psi_\alpha / \partial x_i \partial x_j)(0)$, etc.

3. Consequences of the Theorem

As a corollary of the above expansion theorem, we can characterize certain manifolds by the volume growth of extrinsic balls, when the radius tends to zero. We first consider the case of surfaces in $\mathbb{R}^3$.

PROPOSITION 3.1. *If M^2 is a compact surface in $\mathbb{R}^3$, then $\text{vol}(M_\varepsilon^2(x)) \geq \pi\varepsilon^2 + O(\varepsilon^5)$ for all points $x \in M$ as $\varepsilon \downarrow 0$. Strict inequality obtains for at least one x unless M^2 is a standard sphere (if M is compact) or a plane (if M is non-compact). In either of these exceptional cases, $\text{vol}(M_\varepsilon^2(x)) = \pi\varepsilon^2$ for all x (and ε small enough if M is compact).*

Proof. For $p = 2$ and $N = 3$ we have

$$2\|B\|^2 - \|H\|^2 = 2 \sum_{i=1}^2 \lambda_i^2 - \left(\sum_{i=1}^2 \lambda_i \right)^2 = (\lambda_1 - \lambda_2)^2 \geq 0.$$

This gives the stated inequality for $\text{vol } M_\varepsilon^2(x)$. If equality is obtained, then M^2 is umbilic and hence a sphere if compact and a plane if non-compact. The equality $\text{vol}(M_\varepsilon^2(x)) = \pi\varepsilon^2$ is obvious for the plane; for the sphere of radius r ,

$$M_\varepsilon^2(x) = \{x' \in M^2 : |x - x'| \leq \varepsilon\}$$

corresponds to a region $0 \leq \phi \leq \phi^*$, $0 \leq \theta \leq 2\pi$, and $\rho = r$, in spherical coordinates (ρ, θ, ϕ) . A short calculation shows that $2 - 2r^2 \cos \phi^* = \varepsilon^2$, so that $\text{vol}(M_\varepsilon^2(x)) = \pi\varepsilon^2$.

For $n = \dim M > 2$ we have the following analogue.

PROPOSITION 3.2. *Let M^n be a hypersurface in $\mathbb{R}^{n+1}$ with scalar curvature function K . Then for each $x \in M^n$,*

$$\text{vol}(M_\varepsilon^n(x)) \geq \omega_n \varepsilon^n - [\omega_n (n-2)/4(n+2)(n-1)] K(x) \varepsilon^{n+2} + O(\varepsilon^{n+3}).$$

Strict inequality obtains for at least one point x or M^n is umbilic. Consequently, if $K \equiv n(n-1)/2R^2$ (the scalar curvature of S_R^n) and M^n is compact with

$$\text{vol } M_\varepsilon^n(x) = \omega_n \varepsilon^n - \omega_n [(n-2)/8(n+2)] [n(n+1)/R^2] \varepsilon^{n+2} + O(\varepsilon^{n+3}),$$

then M^n is a standard sphere of radius R . Similarly, if $K(x) \leq 0$ for all x then $\text{vol}(M_\varepsilon^n(x)) \geq \omega_n \varepsilon^n + O(\varepsilon^{n+3})$ with strict inequality for at least one point unless M^n is a hyperplane in which case $\text{vol}(M_\varepsilon^n(x)) = \omega_n \varepsilon^n$ for all x .

Proof. The principal curvatures $\{\lambda_i\}$ satisfy the identity

$$\sum (\lambda_i - \lambda_j)^2 = (n-1) [2 \sum \lambda_i^2 - (\sum \lambda_i)^2] + 2(n-2) K.$$

This yields the stated inequalities when combined with the volume expansion theorem, and equality entails $\sum (\lambda_i - \lambda_j)^2 = 0$, so that M^n is umbilic and hence a sphere.

REMARK. See [6, p. 135] for similar results for the mean exit time.

As a final note, we have:

PROPOSITION 3.3. *If M^p is a minimal submanifold of $\mathbb{R}^N$, then for each $x \in M$, we have*

$$\text{vol}(M_\varepsilon^p(x)) \geq \omega_p \varepsilon^p + \omega_p \|B\|^2 \varepsilon^{p+2}/8(p+2) + O(\varepsilon^{p+3}).$$

Consequently, if for each x

$$\text{vol } M_\varepsilon^p(x) \leq \omega_p \varepsilon^p + O(\varepsilon^{p+3}) \quad (\varepsilon \downarrow 0)$$

then M^p is a p -dimensional plane in $\mathbb{R}^{N+1}$.

The proof is immediate. We note that the inequality

$$\text{vol}(M_\varepsilon^p(x)) \geq \omega_p \varepsilon^p \quad (M^p \text{ minimal})$$

follows from the monotonicity theorem for minimal submanifolds (cf. [9, p. 68; 10, p. 32].

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REMARK. It is not difficult to extend the main result *mutatis mutandis* to submanifolds M^p of an arbitrary Riemannian manifold N . The proof would use normal coordinates in N in place of u and would involve the expansion of the metric tensor in normal coordinates. The coefficient of ε^{p+2} then also involves the curvature of N . We leave the details to the interested reader as an exercise.

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